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JELLY ROLL BATTERY CELL, BATTERY, BATTERY ASSEMBLY AND ELECTRIC DEVICE

Abstract

A jelly roll battery cell, a battery, a battery assembly and an electric device. The jelly roll battery cell includes a main body portion and a tab portion; one side of each positive winding portion in one part of positive winding portions is connected to one positive tab, and one side of each positive winding portion in the other part of the positive winding portions is connected to two positive tabs; and/or, one side of each negative winding portion in one part of negative winding portions is connected to one negative tab, and one side of each negative winding portion in the other part of the negative winding portions is connected to two negative tabs.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of PCT/CN2023/120176, filed on Sep. 20, 2023, which claims priority to Chinese Patent Application No. 202222831093.5, filed on Oct. 26, 2022, the content of each is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] This application relates to the technical field of battery technologies, and in particular to a jelly roll battery cell, a battery, a battery assembly and an electric device.

BACKGROUND

[0003] The background section merely provides background information related to this application and is not necessarily prior art.

[0004] The secondary batteries currently available on the market have attracted extensive attention due to their advantages such as ultra-long service lives, safety in use and large capacities. The battery cell of a secondary battery is a main component that affects the electrochemical performance of the secondary battery. Taking a jelly roll battery cell as an example, which is formed by winding a positive electrode plate, a negative electrode plate and a separator, at least one tab is arranged on the positive electrode plate and the negative electrode plate respectively. In order to increase the flow area and reduce the internal resistance of the battery, for some jelly roll battery cells, a plurality of tabs are arranged on the positive electrode plate and the negative electrode plate respectively. With the trend of increasing capacities of secondary batteries, the number of layers of electrode plates of the jelly roll battery cell has been gradually increased, and the numbers of layers of the plurality of tabs of the positive electrode plate and the plurality of tabs of the negative electrode plate have been increased accordingly, which is prone to the problem of excessive layers of the plurality of tabs of the positive electrode plate and the plurality of tabs of the negative electrode plate. This problem easily causes that existing ultrasonic vibration energy fails to penetrate a plurality of layers of tabs, resulting in the problem of interlayer pseudo soldering between tabs of the battery cell and electrically conductive connecting welds.

SUMMARY

[0005] The technical problem mainly to be solved by this application is the problem of interlayer pseudo soldering between tabs of a battery cell and electrically conductive connecting welds due to excessive layers of tabs of a jelly roll battery cell.

[0006] In order to solve the above technical problem, a technical solution provided by this application is: a jelly roll battery cell, including: [0007] a main body portion, which includes a positive electrode plate, a negative electrode plate and a separator, where the separator is stacked between the positive electrode plate and the negative electrode plate; the main body portion forms a flat winding body by means of winding around a virtual axis; the main body portion includes a plurality of circles of winding portions along a winding direction; the winding portion includes a

positive winding portion, a negative winding portion and a separator winding portion; and [0008] a tab portion, which is connected to one side of the main body portion, and includes a plurality of positive tabs and a plurality of negative tabs, [0009] where one side of each positive winding portion in one part of the positive winding portions is connected to one positive tab, and one side of each positive winding portion in the other part of the positive winding portions is connected to two positive tabs; the two positive tabs of each positive winding portion in the other part of the positive winding portions are located on two opposite sides of the virtual axis along a thickness direction of the main body portion; and/or [0010] one side of each negative winding portion in one part of the negative winding portions is connected to one negative tab, and one side of each negative winding portion in the other part of the negative winding portions is connected to two negative tabs; and the two negative tabs of each negative winding portion in the other part of the negative winding portions are located on two opposite sides of the virtual axis in the thickness direction of the main body portion.

[0011] In some embodiments, one side of each positive winding portion in one part of positive winding portions is connected to one positive tab, and one side of each positive winding portion in the other part of the positive winding portions is connected to two positive tabs; and one side of each negative winding portion in one part of negative winding portions is connected to one negative tab, and one side of each negative winding portion in the other part of the negative winding portions is connected to two negative tabs.

[0012] In some embodiments, one side of each positive winding portion in one part of positive winding portions is connected to one positive tab, one side of each positive winding portion in the other part of the positive winding portions is connected to two positive tabs, and the two negative tabs of each negative winding portion in the other part of the negative winding portions are arranged axially symmetrically along the thickness direction of the main body portion; and each negative winding portion is connected to one negative tab.

[0013] In some embodiments, one side of each negative winding portion in one part of negative winding portions is connected to one negative tab, one side of each negative winding portion in the other part of the negative winding portions is connected to two negative tabs, and the two negative tabs of each negative winding portion in the other part of the negative winding portions are arranged axially symmetrically along the thickness direction of the main body portion; and each positive winding portion is connected to one positive tab.

[0014] In some embodiments, along a direction from inner circles to outer circles, one side of each positive winding portion in the first half of positive winding portions is connected to two positive tabs, and one side of each positive winding portion in the second half of the positive winding portions is connected to one positive tab; and/or

[0015] along the direction from inner circles to outer circles, one side of each negative winding portion in the first half of negative winding portions is connected to two negative tabs, and one side of each negative winding portion in the second half of the negative winding portions is connected to one negative tab.

[0016] Optionally, along the direction from inner circles to outer circles, one side of each positive winding portion in odd-numbered circles of positive winding portions is connected to two positive tabs, and one side of each positive winding portion in even-numbered circles of the positive winding portions is connected to one positive tab; or alternatively, one side of each positive winding portion in odd-numbered circles of positive winding portions is connected to one positive tab, and one side of each positive winding portion in even-numbered circles of the positive winding portions is connected to two positive tabs; and/or

along the direction from inner circles to outer circles, one side of each negative winding portion in odd-numbered circles of negative winding portions is connected to two negative tabs, and one side of each negative winding portion in even-numbered circles of the negative winding portions is connected to one negative tab; or alternatively, one side of each negative winding portion in odd-

numbered circles of negative winding portions is connected to one negative tab, and one side of each negative winding portion in even-numbered circles of the negative winding portions is connected to two negative tabs.

[0017] In some embodiments, the positive electrode plate includes a positive current collector and a positive active material layer arranged on one side of the positive current collector; the negative electrode plate includes a negative current collector and a negative active material layer arranged on one side of the negative current collector;

[0018] one end of the positive tab is connected to the positive current collector of the corresponding positive winding portion, and the other end thereof is away from the positive current collector along the virtual axis; one end of the negative tab is connected to the negative current collector of the corresponding negative winding portion, and the other end thereof is away from the negative current collector along the virtual axis; and along the thickness direction of the main body portion, a plurality of positive tabs located on the same side of the virtual axis are stacked, and a plurality of negative tabs located on the same side of the virtual axis are stacked.

[0019] In some embodiments, the positive tab has the same thickness as the positive current collector; and the negative tab has the same thickness as the negative current collector.

[0020] In some embodiments, in the direction of the virtual axis, the plurality of positive tabs stacked are tabs with an equal length or tabs with gradient lengths; and in the direction of the virtual axis, the plurality of negative tabs stacked are tabs with an equal length or tabs with gradient lengths.

[0021] In some embodiments, the plurality of positive tabs stacked are tabs with an equal length, and the length of the positive tabs is 28%-29% of the width of the positive winding portions connected to the positive tabs; and the plurality of negative tabs stacked are tabs with an equal length, and the length of the negative tabs is 28%-29% of the width of the negative winding portions connected to the negative tabs.

[0022] In some embodiments, the plurality of positive tabs stacked are tabs with gradient lengths; in the plurality of positive tabs, the maximum length of the positive tabs is 28%-29% of the width of the positive winding portions connected to the positive tabs, and the minimum length of the positive tabs is 19%-20% of the width of the positive winding portions connected to the positive tabs; and the lengths of the plurality of positive tabs stacked are decreased progressively layer by layer as per uniform-gradient dimensions along the direction from inner circles to outer circles; and

[0023] the plurality of negative tabs stacked are tabs with gradient lengths; in the plurality of negative tabs, the maximum length of the negative tabs is 28%-29% of the width of the negative winding portions connected to the negative tabs, and the minimum length of the negative tabs is 19%-20% of the width of the negative winding portions connected to the negative tabs; and the lengths of the plurality of negative tabs stacked are decreased progressively layer by layer as per uniform-gradient dimensions along the direction from inner circles to outer circles.

[0024] In some embodiments, the width of the end that is of the positive tab and that is away from the positive current collector is 10%-12.5% of the circumference of the winding portion at the innermost circle of the flat winding body, and the width of the end of the positive tab connected to the positive winding portion is 11%-14% of the circumference of the winding portion at the innermost circle of the flat winding body.

[0025] In some embodiments, the flat winding body has a length direction and a width direction; the virtual axis is parallel to the width direction; and the width of the negative electrode plate is greater than the width of the positive electrode plate, and the length of the negative electrode plate is greater than the length of the positive electrode plate.

[0026] In some embodiments, an innermost circle of the flat winding body is a part of the negative winding portions; and an outermost circle of the flat winding body is a part of the negative winding portions.

[0027] In some embodiments, one side of each positive winding portion in the positive winding

portions is connected to one or two positive tabs; one side of each negative winding portion in the negative winding portions is connected to one or two negative tabs; or
[0028] one side of each positive winding portion in the positive winding portions is connected to one or two positive tabs; in the negative winding portions, the negative winding portion at the innermost circle or the negative winding portion at the outermost circle is not provided with the negative tab; and one side of each of the other negative winding portions is connected to one or two negative tabs.

[0029] In order to solve the above technical problem, another technical solution provided by this application is: a battery, including any one of the above jelly roll battery cells.

[0030] In order to solve the above technical problem, yet another technical solution provided by this application is: a battery assembly, including the above batteries.

[0031] In order to solve the above technical problem, still yet another technical solution provided by this application is: an electric device, including the above battery assembly.

[0032] Beneficial effects: in this application, by connecting one positive/negative tab to one side of each positive/negative winding portion in one part of positive/negative winding portions, and connecting two positive/negative tabs to one side of each positive/negative winding portion in the other part of the positive/negative winding portions, the number of positive/negative tabs is increased, and the flow area of the jelly roll battery cell is increased; in addition, due to a restriction on the number of positive/negative tabs, interlayer pseudo soldering between excessive positive/negative tabs and electrically conductive connecting welds is avoided.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0033] In order to explain the technical solutions in the embodiments of this application more clearly, drawings to be used in the description of embodiments will be described briefly below. Apparently, the accompanying drawings in the following description show merely some embodiments of this application, and other drawings can be derived from these accompanying drawings by a person of ordinary skill in the art without making any creative efforts.

[0034] FIG. 1 is a structural view of a jelly roll battery cell according to this application;

[0035] FIG. 2 is a top view of the jelly roll battery cell shown in FIG. 1;

[0036] FIG. 3 is a structural view of electrode plates and a separator of the jelly roll battery cell shown in FIG. 1;

[0037] FIG. 4a is a partial sectional view along line A-A of the jelly roll battery cell shown in FIG. 2;

[0038] FIG. 4b is a partial sectional view along line B-B of the jelly roll battery cell shown in FIG. 2;

[0039] FIG. 5a is a second partial sectional view along line A-A of the jelly roll battery cell shown in FIG. 2;

[0040] FIG. 5b is a second partial sectional view along line B-B of the jelly roll battery cell shown in FIG. 2;

[0041] FIG. 6 is a top view of a second embodiment of a jelly roll battery cell according to this application;

[0042] FIG. 7 is a top view of a third embodiment of a jelly roll battery cell according to this application;

[0043] FIG. 8 is a top view of a fourth embodiment of a jelly roll battery cell according to this application;

[0044] FIG. 9 is a top view of a fifth embodiment of a jelly roll battery cell according to this application;

[0045] FIG. **10** is a top view of a sixth embodiment of a jelly roll battery cell according to this application;

[0046] FIG. **11** is a structural view of a battery according to this application;

[0047] FIG. **12** is a structural view of a battery assembly according to this application; and

[0048] FIG. **13** is a structural view of an electric device according to this application.

REFERENCE NUMERALS

[0049] **100**—jelly roll battery cell;

[0050] **10**—main body portion;

[0051] **11**—positive electrode plate;

[0052] **12**—negative electrode plate;

[0053] **13**—separator;

[0054] Y—virtual axis;

[0055] H—thickness of main body portion;

[0056] X—axis perpendicular to thickness direction of main body portion and perpendicular to direction of virtual axis;

[0057] **101**—winding portion;

[0058] **101a**—positive winding portion;

[0059] **101b**—negative winding portion;

[0060] **101c**—separator winding portion;

[0061] **20**—tab portion;

[0062] **21**—positive tab;

[0063] **22**—negative tab;

[0064] **11a**—positive current collector;

[0065] **11b**—positive active material layer;

[0066] **12a**—negative current collector;

[0067] **12b**—negative active material layer;

[0068] L—length of positive/negative tab;

[0069] W—width of positive/negative winding portion;

[0070] **K1**—width of end of positive/negative tab away from positive/negative current collector;

[0071] **K2**—width of end of positive/negative tab connected to positive/negative current collector;

[0072] **S1**—first frame;

[0073] **S2**—second frame;

[0074] **200**—battery;

[0075] **210**—shell;

[0076] **300**—battery assembly;

[0077] **400**—electric vehicle;

[0078] **401**—controller; and

[0079] **402**—motor.

DETAILED DESCRIPTION OF EMBODIMENTS

[0080] The following will clearly and completely describe the technical solutions in embodiments of this application in combination with the drawings in the embodiments of this application.

Obviously, the described embodiments are merely a part of the embodiments of this application, not all of the embodiments. All other embodiments obtained by a person of ordinary skill in the art based on the embodiments of this application without making any creative effort fall within the protection scope of this application.

[0081] In this application, terms “first”, “second” and “third” are used for descriptive purposes only, and cannot be understood as indicating or implying relative importance, or implicitly specifying quantities of technical features indicated. Thus, features defined with “first”, “second” and “third” can expressly or implicitly include at least one of the features. In the description of this application, the meaning of “a plurality of” is at least two, such as two or three, unless otherwise

clearly and specifically limited. All directional indications (such as up, down, left, right, front and rear) in embodiments of this application are only used to explain relative positional relations and motions among various components under a certain specific attitude (as shown in drawings), and correspondingly vary with the specific attitude. In addition, terms “include” and “have” and their variants in any forms are intended to cover non-exclusive inclusion. For example, processes, methods, systems, products or apparatuses including a series of steps or units are not limited to listed steps or units, and optionally further include non-listed steps or units, or optionally further include other steps or units that are inherent to these processes, methods, products or apparatuses. [0082] Reference to “embodiments” in this specification means that particular features, structures, or characteristics described with reference to the embodiments may be included in at least one embodiment of this application. Occurrence of the phrase in various places in this specification does not necessarily refer to the same embodiment, nor is an independent or alternative embodiment that is mutually exclusive with other embodiments. It is understood explicitly and implicitly by a person skilled in the art that the embodiments described in this specification may be combined with other embodiments.

[0083] Researcher of this application have found through research that a flow area of a battery cell is the product of the sum of thicknesses of a plurality of layers of tabs and a width of tabs. For some battery cell designs with extreme energy densities, substrates should be as thin as possible. In order to meet requirements for the flow area, a number of tab layers or a bottom width of tabs has to be increased. When the number of tab layers remains unchanged and the width of tabs is increased, in case that excursions of tabs conform to specifications, an excursion to the left may interfere with a lower plastic piece under an explosion-proof valve, leading to the risk of tabs being adversely lap-joined to the lower plastic piece (under a high-temperature cycling condition of the battery cell, the lower plastic piece softens and deforms, and tabs lap-joined to the lower plastic piece may directly come into contact with the explosion-proof valve of a battery, which causes short circuits and greatly increases the risk of fire in the battery); alternatively, an excursion to the right may interfere with a liquid injection hole, causing the risk of clogging of the liquid injection hole. Therefore, increasing the number of tab layers is an optional way to meet requirements for the flow area. However, the number of tab layers cannot be increased without limits. In case that the number of tab layers is increased, it will be necessary to avoid the problem of interlayer pseudo soldering between tabs of the battery cell and electrically conductive connecting welds.

[0084] This application will be described in detail below in conjunction with drawings and embodiments.

[0085] Referring to FIGS. 1 to 4b, FIG. 1 is a structural view of a jelly roll battery cell provided in this application, FIG. 2 is a top view of the jelly roll battery cell provided in FIG. 1, FIG. 3 is a structural view of electrode plates and a separator of the jelly roll battery cell provided in FIG. 1, FIG. 4a is a partial sectional view along line A-A of the jelly roll battery cell provided in FIG. 2, and FIG. 4b is a partial sectional view along line B-B of the jelly roll battery cell provided in FIG. 2.

[0086] Referring to FIGS. 1 to 4b, provided in the embodiments of this application is a jelly roll battery cell **100**, including a main body portion **10** and a tab portion **20**,

[0087] where the main body portion **10** includes a positive electrode plate **11**, a negative electrode plate **12** and a separator **13**. The separator **13** is stacked between the positive electrode plate **11** and the negative electrode plate **12**. The main body portion **10** forms a flat winding body by means of winding around a virtual axis Y. The main body portion **10** includes a plurality of circles of winding portions **101** along a winding direction. The winding portion **101** includes a positive winding portion **101a**, a negative winding portion **101b** and a separator winding portion **101c**. Each winding portion **101** makes a revolution around the virtual axis Y, each winding portion **101** includes two layers of winding sub-portions in a thickness direction H of the main body portion **10**, and each layer of winding sub-portions makes a semi-revolution around the virtual axis Y. The two layers of

winding sub-portions of each winding portion **101** are arranged axially symmetrically in a thickness direction of the main body portion **10**, and namely, are arranged axially symmetrically along an axis X perpendicular to the thickness direction H of the main body portion **10** and perpendicular to a direction of the virtual axis Y. Each layer of winding sub-portions includes a positive winding sub-portion, a separator winding sub-portion and a negative winding sub-portion, which are stacked.

[0088] The tab portion **20** is connected to one side of the main body portion **10**, and includes a plurality of positive tabs **21** and a plurality of negative tabs **22**.

[0089] One side of each positive winding portion **101a** in one part of positive winding portions **101a** is connected to one positive tab **21**, and one side of each positive winding portion **101a** in the other part of the positive winding portions **101a** is connected to two positive tabs **21**. The two positive tabs **21** of each positive winding portion **101a** in the other part of the positive winding portions **101a** are located on two opposite sides of the virtual axis Y along the thickness direction H of the main body portion **10**. That is, one side of each positive winding portion **101a** in one part of positive winding portions **101a** is connected to one positive tab **21**, and one side of each layer of positive winding sub-portions of each positive winding portion **101a** in the other part of the positive winding portions **101a** is connected to one positive tab **21**.

[0090] And/or, one side of each negative winding portion **101b** in one part of negative winding portions **101b** is connected to one negative tab **22**, and one side of each negative winding portion **101b** in the other part of the negative winding portions **101b** is connected to two negative tabs **22**. The two negative tabs **22** of each negative winding portion **101b** in the other part of the negative winding portions **101b** are located on two opposite sides of a virtual axis Y along the thickness direction of the main body portion **10**. That is, one side of each negative winding portion **101b** in one part of negative winding portions **101b** is connected to one negative tab **22**, and one side of each negative winding portion **101b** in the other part of the negative winding portions **101b** is connected to one negative tab **22**.

[0091] In this application:

[0092] the jelly roll battery cell **100** represents a flat winding body formed by winding the positive electrode plate **11**, the negative electrode plate **12** and the separator **13** around the virtual axis Y by hand or by machine. Specifically, in a procedure for forming the battery cell, the electrode plates and the separator **13** provide winding support for winding the battery cell by using a winding needle as a supporting component of the battery cell, and the battery cell is wound on an outer side of the winding needle. After the step of winding the battery cell is completed, the winding needle can be removed from the battery cell.

[0093] The main body portion **10** represents a flat winding body formed by winding the positive electrode plate **11**, the negative electrode plate **12** and the separator **13**. The positive electrode plate **11** includes a positive current collector **11a** and a positive active material layer **11b**. A surface of the positive current collector **11a** is coated with the positive active material layer **11b**. Taking a lithium-ion battery as an example, a material of the positive current collector **11a** may be aluminum, and a positive active material may be lithium cobaltate, lithium iron phosphate, ternary lithium or lithium manganate. The negative electrode plate **12** includes a negative current collector **12a** and a negative active material layer **12b**. A surface of the negative current collector **12a** is coated with the negative active material layer **12b**. Taking the lithium-ion battery as an example, a material of the negative current collector **12a** may be copper, and a negative active material may be carbon or silicon, etc. A material of the separator **13** may be PP or PE. The virtual axis Y represents a line connecting centroids of all cross-sections of the main body portion **10**.

[0094] The winding portion **101** represents a part of the positive electrode plate **11**, a part of the negative electrode plate **12** or a part of the separator **13** of each circle of winding in the main body portion **10** formed by a plurality of circles of winding. The positive winding portion **101a** represents the part of the positive electrode plate **11** of each circle of winding in the main body

portion **10** formed by the plurality of circles of winding. The negative winding portion **101b** represents the part of the negative electrode plate **12** of each circle of winding in the main body portion **10** formed by the plurality of circles of winding. The separator winding portion **101c** represents the part of the separator **13** of each circle of winding in the main body portion **10** formed by the plurality of circles of winding.

[0095] The tab portion **20** is connected to the main body portion **10**, and is an electrical conductor for leading out the positive electrode plate **11** and the negative electrode plate **12** of the main body portion **10**. The positive tab **21** represents a tab for leading out the positive electrode plate **11**, and can be integrally formed with the positive current collector **11a**. The negative tab **22** represents a tab for leading out the negative electrode plate **12**, and can be integrally formed with the negative current collector **12a**. In some specific embodiments, the positive current collector **11a** uncoated with the positive active material protrudes relative to the positive current collector **11a** coated with the positive active material, and a part of the positive current collector **11a** uncoated with the positive active material serves as the positive tab **21**. The negative current collector **12a** uncoated with the negative active material protrudes relative to the negative current collector **12a** coated with the negative active material, and a part of the negative current collector **12a** uncoated with the negative active material serves as the negative tab **22**.

[0096] One positive tab **21** is connected to one side of the positive winding portion **101a** means that one positive tab **21** is disposed on one side of the positive winding portion **101a** along the virtual axis Y. Two positive tabs **21** are connected to one side of the positive winding portion **101a** means that two positive tabs **21** are disposed on one side of the positive winding portion **101a** along the virtual axis Y. One negative tab **22** is connected to one side of the negative winding portion **101b** means that one negative tab **22** is disposed on one side of the negative winding portion **101b** along the virtual axis Y. Two negative tabs **22** are connected to one side of the negative winding portion **101b** means that two negative tabs **22** are disposed on one side of the negative winding portion **101b** along the virtual axis Y. Connecting or disposing one positive tab **21** means that only one positive tab **21** is connected or disposed. Connecting or disposing two positive tabs **21** means that only two positive tabs **21** are connected or disposed. Connecting or disposing one negative tab **22** means that only one negative tab **22** is connected or disposed. Connecting or disposing two negative tabs **22** means that only two negative tabs **22** are connected or disposed.

[0097] In this application, by connecting one positive tab **21** to one side of each positive winding portion **101a** in one part of positive winding portions **101a**, and connecting two positive tabs **21** to one side of each positive winding portion **101a** in the other part of the positive winding portions **101a**, the number of positive tabs **21** is increased, and the flow area of the jelly roll battery cell **100** is increased; in addition, due to a restriction on the number of positive tabs **21**, interlayer pseudo soldering between excessive positive tabs **21** and electrically conductive connecting welds is avoided.

[0098] In this application, by connecting one negative tab **22** to one side of each negative winding portion **101b** in one part of negative winding portions **101b**, and connecting two negative tabs **22** to one side of each negative winding portion **101b** in the other part of the negative winding portions **101b**, the number of negative tabs **22** is increased, and the flow area of the jelly roll battery cell **100** is increased; in addition, due to a restriction on the number of negative tabs **22**, interlayer pseudo soldering between excessive negative tabs **22** and electrically conductive connecting welds is avoided.

[0099] Specifically, as shown in FIG. 2, in this embodiment, one side of each negative winding portion **101b** in one part of negative winding portions **101b** is connected to one negative tab **22**, one side of each negative winding portion **101b** in the other part of the negative winding portions **101b** is connected to two negative tabs **22**, and the two negative tabs **22** of each negative winding portion **101b** in the other part of the negative winding portions **101b** are arranged axially symmetrically along the thickness direction of the main body portion **10**; and each positive winding

portion **101a** is connected to one positive tab **21**. That is, a plurality of negative tabs **22** in a first frame **S1** and a plurality of negative tabs **22** in a second frame **S2** in FIG. 2 are arranged axially symmetrically along the axis **X** perpendicular to the thickness direction of the main body portion **10** and perpendicular to the direction of the virtual axis **Y**.

[0100] In this embodiment, the number of the plurality of negative tabs **22** is increased than that when each circle of the negative winding portion **101b** is provided with only one negative tab **22**, which increases the flow area of the jelly roll battery cell **100** and reduces the internal resistance of the jelly roll battery cell **100**, and the number of the plurality of negative tabs **22** does not lead to the problem of interlayer pseudo soldering between excessive negative tabs **22** and electrically conductive connecting welds. For example, the situation that each circle of the negative winding portion **101b** is provided with two negative tabs **22**, resulting in the problem of interlayer pseudo soldering of electrically conductive connecting welds is avoided.

[0101] The number of circles of the negative winding portions **101b** provided with two negative tabs **22** can be specifically selected and designed according to the total number of circles of the negative winding portions **101b**.

[0102] Continuously referring to FIGS. 1-3, in the jelly roll battery cell **100**, the positive electrode plate **11** includes a positive current collector **11a** and a positive active material layer **11b** arranged on one side of the positive current collector **11a**; the negative electrode plate **12** includes a negative current collector **12a** and a negative active material layer **12b** arranged on one side of the negative current collector **12a**.

[0103] One end of the positive tab **21** is connected to the positive current collector **11a** of the corresponding positive winding portion **101a**, and the other end thereof is away from the positive current collector **11a** along the virtual axis **Y**. One end of the negative tab **22** is connected to the negative current collector **12a** of the corresponding negative winding portion **101b**, and the other end thereof is away from the negative current collector **12a** along the virtual axis **Y**. Along the thickness direction of the main body portion **10**, a plurality of positive tabs **21** located on the same side of the virtual axis **Y** are stacked, and a plurality of negative tabs **22** located on the same side of the virtual axis **Y** are stacked.

[0104] In some specific embodiments, the positive tab **21** has the same thickness as the positive current collector **11a**; and the negative tab **22** has the same thickness as the negative current collector **12a**.

[0105] In some embodiments, as shown in FIG. 2, an innermost circle of the flat winding body is a part of the negative winding portions **101b**; and an outermost circle of the flat winding body is a part of the negative winding portions **101b**, which is intended to slow down the occurrence of lithium precipitation. In other words, the negative winding portion **101b** has at least one more circle than the positive winding portion **101a**.

[0106] In some embodiments, in the direction of the virtual axis **Y**, the plurality of positive tabs **21** stacked are tabs with an equal length or tabs with gradient lengths. In the direction of the virtual axis **Y**, the plurality of negative tabs **22** stacked are tabs with an equal length or tabs with gradient lengths. In the thickness direction **H** of the main body portion **10**, the plurality of positive tabs **21**/negative tabs **22** stacked can overlap or partially overlap (i.e., be staggered) or be arranged at intervals in respective width directions thereof. It can be understood that by means of a solution in which the plurality of positive tabs **21**/negative tabs **22** stacked partially overlap or are arranged at intervals in the thickness direction **H** of the main body portion **10**, the widths of the plurality of positive tabs **21**/negative tabs **22** are increased, and the areas of a copper adapter and an aluminum adapter are also increased accordingly. In case of an excessively narrow design in a width direction of the battery cell, there will be problems of interference between negative and positive electrodes or interference between the liquid injection hole and a positive adapter, in which the former has the risk of short circuits, and the latter affects liquid injection of the battery. Therefore, in this embodiment, the plurality of positive tabs **21**/negative tabs **22** stacked overlap in respective width

directions thereof.

[0107] In some embodiments, as shown in FIG. 3 and FIG. 4a, in the direction of the virtual axis Y, the plurality of positive tabs **21** stacked are tabs with an equal length, and the length L of the positive tabs **21** is 28%-29% of the width W of the positive winding portions **101a** connected to the positive tabs **21**. As shown in FIG. 3 and FIG. 4b, in the direction of the virtual axis Y, the plurality of negative tabs **22** stacked are tabs with an equal length, and the length L of the negative tabs **22** is 28%-29% of the width W of the negative winding portions **101b** connected to the negative tabs **22**.

[0108] In some embodiments, as shown in FIG. 3, the width K1 of the end that is of the positive tab **21** and that is away from the positive current collector **11a** is 10%-12.5% of the circumference of the winding portion **101** at the innermost circle of the flat winding body, and the width K2 of the end that is of the positive tab **21** and that is connected to the positive current collector **11a** is 11%-14% of the circumference of the winding portion **101** at the innermost circle of the flat winding body. In some embodiments, the flat winding body has a length direction and a width direction; the length direction is a length direction of the flat winding body before winding along the winding direction or after unwinding; the virtual axis Y is parallel to the width direction; and the width of the negative electrode plate **12** is greater than the width of the positive electrode plate **11**, and the length of the negative electrode plate **12** is greater than the length of the positive electrode plate **11**.

[0109] In some embodiments, one side of each positive winding portion **101a** in the

[0110] positive winding portions **101a** is connected to the positive tab **21**, and is specifically connected to one or two positive tabs **21**. One side of each negative winding portion **101b** in the negative winding portions **101b** is connected to the negative tab **22**, and is specifically connected to one or two negative tabs **22**. Alternatively, one side of each positive winding portion **101a** in the positive winding portions **101a** is connected to the positive tab **21**, and is specifically connected to one or two positive tabs **21**; in the negative winding portions **101b**, the negative winding portion **101b** at the innermost circle or the negative winding portion **101b** at the outermost circle is not provided with the negative tab **22**; and one side of each of the other negative winding portions **101b** is connected to the negative tab **22**, and is specifically connected to one or two negative tabs **22**.

[0111] In the jelly roll battery cell **100** in some embodiments, along a direction from inner circles to outer circles, one side of each positive winding portion **101a** in the first half of positive winding portions **101a** is connected to two positive tabs **21**, and one side of each positive winding portion **101a** in the second half of the positive winding portions **101a** is connected to one positive tab **21** (as shown in FIG. 6 and FIG. 7).

[0112] And/or, along the direction from inner circles to outer circles, one side of each negative winding portion **101b** in the first half of negative winding portions **101b** is connected to two negative tabs **22**, and one side of each negative winding portion **101b** in the second half of the negative winding portions **101b** is connected to one negative tab **22** (as shown in FIG. 2 and FIG. 7).

[0113] Specifically, by connecting two positive tabs **21** to the inner circles of the positive winding portions **101a** in the jelly roll battery cell **100**, and connecting one positive tab **21** to the outer circles of the positive winding portions **101a**, the flow area of the jelly roll battery cell **100** is increased, and the internal resistance of the jelly roll battery cell **100** is decreased; the number of the plurality of positive tabs **21** does not lead to the problem of interlayer pseudo soldering between excessive positive tabs **21** and electrically conductive connecting welds, and the arrangement mode of the plurality of positive tabs **21** causes the positive tabs **21** to be closely spaced, such that the welding effect is good. By connecting two negative tabs to one side of each negative winding portion in the jelly roll battery cell **100**, and connecting one negative tab to one side of the outer circles of the negative winding portions **101b**, the flow area of the jelly roll battery cell **100** is increased, and the internal resistance of the jelly roll battery cell **100** is reduced; the number of the plurality of negative tabs **22** does not lead to the problem of interlayer pseudo soldering between

excessive negative tabs **22** and electrically conductive connecting welds, and the arrangement mode of the plurality of negative tabs **22** causes the negative tabs **22** to be closely spaced, such that the welding effect is good.

[0114] Referring to FIGS. **5a-5b**, FIG. **5a** is a second partial sectional view along line A-A of the jelly roll battery cell provided in FIG. **2**, and FIG. **5b** is a second partial sectional view along line B-B of the jelly roll battery cell provided in FIG. **2**.

[0115] Referring to FIGS. **5a** and **5b**, the jelly roll battery cell **100** provided in this embodiment differs from the jelly roll battery cell **100** provided in FIG. **4a** in that: as shown in FIG. **5a**, in the direction of the virtual axis Y, the plurality of positive tabs **21** in this embodiment are tabs with gradient lengths; in the plurality of positive tabs **21**, the maximum length of the positive tabs **21** is 28%-29% of the width W of the positive winding portions **101a** connected to the positive tabs **21**, and the minimum length of the positive tabs **21** is 19%-20% of the width W of the positive winding portions **101a** connected to the positive tabs **21**; and the lengths L of the plurality of positive tabs **21** stacked are decreased progressively layer by layer as per uniform-gradient dimensions along the direction from inner circles to outer circles.

[0116] As shown in FIG. **5b**, the plurality of negative tabs **22** stacked are tabs with gradient lengths; in the plurality of negative tabs **22**, the maximum length of the negative tabs **22** is 28%-29% of the width W of the negative winding portions **101b** connected to the negative tabs **22**, and the minimum length of the negative tabs **22** is 19%-20% of the width W of the negative winding portions **101b** connected to the negative tabs **22**; and the lengths L of the plurality of negative tabs **22** stacked are decreased progressively layer by layer as per uniform-gradient dimensions along the direction from inner circles to outer circles.

[0117] Referring to FIG. **6**, FIG. **6** is a top view of a second embodiment of a jelly roll battery cell provided in this application.

[0118] Referring to FIG. **6**, the jelly roll battery cell **100** provided in this embodiment differs from the jelly roll battery cell **100** provided in FIG. **2** in that: in this embodiment, one side of each positive winding portion **101a** in one part of positive winding portions **101a** is connected to one positive tab **21**, and one side of each positive winding portion **101a** in the other part of the positive winding portions **101a** is connected to two positive tabs **21**, and the two positive tabs **21** of each positive winding portion **101a** in the other part of the positive winding portions **101a** are arranged axially symmetrically along the thickness direction H of the main body portion **10**; and each negative winding portion **101b** is connected to one negative tab **22**.

[0119] In this embodiment, the number of the plurality of positive tabs **21** is increased than that when each circle of the positive winding portion **101a** is provided with only one positive tab **21**, which increases the flow area of the jelly roll battery cell **100** and reduces the internal resistance of the jelly roll battery cell **100**, and the number of the plurality of positive tabs **21** does not lead to the problem of interlayer pseudo soldering between excessive positive tabs **21** and electrically conductive connecting welds.

[0120] Referring to FIG. **7**, FIG. **7** is a top view of a third embodiment of a jelly roll battery cell provided in this application.

[0121] Referring to FIG. **7**, the jelly roll battery cell **100** shown in this embodiment differs from the jelly roll battery cell **100** shown in FIG. **2** in that: in the jelly roll battery cell **100** provided in this embodiment, one side of each positive winding portion **101a** in one part of positive winding portions **101a** is connected to one positive tab **21**, and one side of each positive winding portion **101a** in the other part of the positive winding portions **101a** is connected to two positive tabs **21**. Additionally, one side of each negative winding portion **101b** in one part of negative winding portions **101b** is connected to one negative tab **22**, and one side of each negative winding portion **101b** in the other part of the negative winding portions **101b** is connected to two negative tabs **22**.

[0122] In the above embodiment, the number of the plurality of positive tabs **21** and the number of the plurality of negative tabs **22** are both increased, which jointly improve the overall flow area of

the jelly roll battery cell **100**, and avoid interlayer pseudo soldering between excessive positive tabs **21**, excessive negative tabs **22** and electrically conductive connecting welds, and the number of the plurality of negative tabs **21** has the same increasing trend as that of the plurality of positive tabs **22**, which has a synergistic effect on the improvement in the overall flow ability of the jelly roll battery cell **100**.

[0123] In some embodiments, along the direction from inner circles to outer circles, one side of each positive winding portion **101a** in odd-numbered circles of positive winding portions **101a** is connected to two positive tabs **21**, and one side of each positive winding portion **101a** in even-numbered circles of the positive winding portions **101a** is connected to one positive tab **21**.

Alternatively, one side of each positive winding portion **101a** in odd-numbered circles of positive winding portions **101a** is connected to one positive tab **21**, and one side of each positive winding portion **101a** in even-numbered circles of positive winding portions **101a** is connected to two positive tabs **21**. And/or, along the direction from inner circles to outer circles, one side of each negative winding portion **101b** in odd-numbered circles of negative winding portions **101b** is connected to two negative tabs **22**, and one side of each negative winding portion **101b** in even-numbered circles of the negative winding portions **101b** is connected to one negative tab **22**; alternatively, one side of each negative winding portion **101b** in odd-numbered circles of negative winding portions **101b** is connected to one negative tab **22**, and one side of each negative winding portion **101b** in even-numbered circles of the negative winding portions **101b** is connected to two negative tabs **22**.

[0124] Referring to FIG. **8**, FIG. **8** is a top view of a fourth embodiment of the jelly roll battery cell provided in this application.

[0125] Referring to FIG. **8**, the jelly roll battery cell **100** shown in this embodiment differs from the jelly roll battery cell **100** shown in FIG. **2** in that: in this embodiment, along the direction from inner circles to outer circles, one side of each positive winding portion **101a** in odd-numbered circles of positive winding portions **101a** is connected to two positive tabs **21**, and one side of each positive winding portion **101a** in even-numbered circles of the positive winding portions **101a** is connected to one positive tab **21**; and one side of each circle of the negative winding portion **101b** is connected to one negative tab **22**. Such an arrangement mode not only increases the flow area of the jelly roll battery cell **100**, but also balances distribution of the plurality of positive tabs **21** and improves the welding effect of the plurality of positive tabs **21**.

[0126] Referring to FIG. **9**, FIG. **9** is a top view of a fifth embodiment of a jelly roll battery cell provided in this application.

[0127] Referring to FIG. **9**, the jelly roll battery cell **100** shown in this embodiment differs from the jelly roll battery cell **100** shown in FIG. **2** in that: in this embodiment, along the direction from inner circles to outer circles, one side of each negative winding portion **101b** in odd-numbered circles of positive winding portions **101b** is connected to two negative tabs **22**, and one side of each negative winding portion **101b** in even-numbered circles of the negative winding portions **101b** is connected to one negative tab **22**; and one side of each circle of the positive winding portion **101b** is connected to one positive tab **21**. Such an arrangement mode not only increases the flow area of the jelly roll battery cell **100**, but also balances distribution of the plurality of negative tabs **22** and improves the welding effect of the plurality of negative tabs **22**.

[0128] Referring to FIG. **10**, FIG. **10** is a top view of a sixth embodiment of a jelly roll battery cell provided in this application.

[0129] Referring to FIG. **10**, the jelly roll battery cell **100** shown in this embodiment differs from the jelly roll battery cell **100** shown in FIG. **2**: in this embodiment, along the direction from inner circles to outer circles, one side of each negative winding portion **101b** in odd-numbered circles of negative winding portions **101b** is connected to two negative tabs **22**, one side of each negative winding portion **101b** in even-numbered circles of the negative winding portions **101b** is connected to one negative tab **22**. Additionally, along the direction from inner circles to outer circles, one side

of each positive winding portion **101a** in odd-numbered circles of positive winding portions **101a** is connected to two positive tabs **21**, one side of each positive winding portion **101a** in even-numbered circles of the positive winding portions **101a** is connected to one positive tab **21**, and one side of each circle of the negative winding portion **101b** is connected to one negative tab **22**. Such an arrangement mode not only increases the flow area of the jelly roll battery cell **100**, but also balances distribution of the plurality of positive tabs **21** and the plurality of negative tabs **22** and improves the welding effect of the plurality of positive tabs **21** and the plurality of negative tabs **22**. [0130] Referring to FIG. **11**, FIG. **11** is a structural view of a battery provided in this application. [0131] Referring to FIG. **11**, provided in this application is a battery **200**, including a shell **210** and any one of above jelly roll battery cells **100**.

[0132] In this embodiment, the battery **200** represents a single physical module, which includes one or more jelly roll battery cells **100** (as shown in FIGS. **1-10**) to improve the voltage and capacity. The jelly roll battery cell **100** includes a main body portion **10** and a tab portion **20**, where the main body portion **10** includes a positive electrode plate **11**, a negative electrode plate **12** and a separator **13**. The separator **13** is stacked between the positive electrode plate **11** and the negative electrode plate **12**. The main body portion **10** forms a flat winding body by means of winding around a virtual axis Y. The main body portion **10** includes a plurality of circles of winding portions **101** along a winding direction. The winding portion **101** includes a positive winding portion **101a**, a negative winding portion **101b** and a separator winding portion **101c**. The tab portion **20** is connected to one side of the main body portion **10**, and includes a plurality of positive tabs **21** and a plurality of negative tabs **22**.

[0133] One side of each positive winding portion **101a** in one part of positive winding portions **101a** is connected to one positive tab **21**, and one side of each positive winding portion **101a** in the other part of the positive winding portions **101a** is connected to two positive tabs **21**. The two positive tabs **21** of each positive winding portion **101a** in the other part of the positive winding portions **101a** are located on two opposite sides of the virtual axis Y along the thickness direction H of the main body portion **10**.

[0134] And/or, one side of each negative winding portion **101b** in one part of negative winding portions **101b** is connected to one negative tab **22**, and one side of each negative winding portion **101b** in the other part of the negative winding portions **101b** is connected to two negative tabs **22**. The two negative tabs **22** of each negative winding portion **101b** in the other part of the negative winding portions **101b** are located on two opposite sides of a virtual axis Y along the thickness direction of the main body portion **10**.

[0135] Referring to FIG. **12**, FIG. **12** is a structural view of a battery assembly provided in this application.

[0136] Referring to FIG. **12**, provided in this application is a battery assembly **300**, including the above batteries **200**. The battery assembly **300** may further include a box, which is used for providing an accommodating space for the batteries **200**, and may be of various shapes. In the battery assembly **300**, a plurality of batteries **200** may be connected in series or in parallel, or in a series-parallel manner, which means that both series connection and parallel connection are involved in the plurality of batteries **200**. The plurality of batteries **200** may be directly connected in series or in parallel or in the series-parallel manner, and then the plurality of batteries **200** are accommodated in the box as a whole; and of course, it is also possible for the battery assembly **300** that the plurality of batteries **200** are firstly connected in series or in parallel or in the series-parallel manner to form battery modules, and then a plurality of battery modules are connected in series or in parallel or in the series-parallel manner as a whole, and are accommodated in the box. The battery assembly **300** may further include other structures. For example, the battery assembly **300** may further include a busbar component for realizing electrical connections among the plurality of batteries **200**. Each battery **200** may be a secondary battery or a primary battery; and same may also be a lithium-sulfur battery, a sodium-ion battery or a magnesium-ion battery, but is not limited

thereto.

[0137] Referring to FIG. 13, FIG. 13 is a structural view of an electric device provided in this application.

[0138] Referring to FIG. 13, provided in this application is an electric device, including the above battery assembly 300.

[0139] The electric device may be a mobile phone, a computer, an electric motorcycle, an electric vehicle or the like. This embodiment takes the electric vehicle 400 as an example for description. The battery assembly 300 is arranged in the electric vehicle 400, and the battery assembly 300 may be arranged at the bottom, front or rear of the electric vehicle 400. The battery assembly 300 may be used for power supply of the electric vehicle 400; for example, the battery assembly 300 may serve as an operational power supply of the electric vehicle 400. The electric vehicle 400 may further include a controller 401 and a motor 402, and the controller 401 is used for controlling the battery assembly 300 to supply the motor 402 with power, such as operation power for the electric vehicle 400 during startup, navigation and driving.

[0140] In some embodiments of this application, the battery assembly 300 may serve as not only an operational power supply of the electric vehicle 400, but also a driving power supply of the electric vehicle 400 to supply driving power for the electric vehicle 400.

[0141] In several embodiments provided in this application, it should be understood that systems, devices and methods disclosed can be implemented by other means. For example, embodiments of the device described above are only illustrative; for example, division of units is only division of logical functions, and other division methods can be used for actual implementation; for example, a plurality of units or assemblies can be combined or integrated into another system; alternatively, some features can be omitted or are not implemented. Moreover, mutual coupling or direct coupling or communication connection shown or discussed can be achieved by means of some interfaces, and indirect coupling or communication connection of devices or units can be in electrical, mechanical or other forms.

[0142] In addition, various functional units in embodiments of this application can be integrated into one processing unit; alternatively, each unit can exist separately and physically, or two or more units can be integrated into one unit. The integrated units can be implemented as hardware or functional units of software.

[0143] The foregoing are only embodiments of this application and do not limit the patent scope of this application. All equivalent structure or equivalent workflow transformations, which are made by means of contents of the description and drawings of this application and directly or indirectly applied in other relevant technical fields, are similarly included in the patent protection scope of this application.

Claims

1. A jelly roll battery cell, comprising: a main body portion, which comprises a positive electrode plate, a negative electrode plate and a separator, wherein the separator is stacked between the positive electrode plate and the negative electrode plate; the main body portion forms a flat winding body by means of winding around a virtual axis; the main body portion comprises a plurality of circles of winding portions along a winding direction; the winding portion comprises a positive winding portion, a negative winding portion and a separator winding portion; and a tab portion, which is connected to one side of the main body portion, and comprises a plurality of positive tabs and a plurality of negative tabs, wherein one side of each positive winding portion in one part of positive winding portions is connected to one positive tab, and one side of each positive winding portion in the other part of the positive winding portions is connected to two positive tabs; the two positive tabs of each positive winding portion in the other part of the positive winding portions are located on two opposite sides of the virtual axis along a thickness direction of the main

body portion; and/or one side of each negative winding portion in one part of negative winding portions is connected to one negative tab, and one side of each negative winding portion in the other part of the negative winding portions is connected to two negative tabs; and the two negative tabs of each negative winding portion in the other part of the negative winding portions are located on two opposite sides of the virtual axis along the thickness direction of the main body portion.

2. The jelly roll battery cell according to claim 1, wherein one side of each positive winding portion in one part of positive winding portions is connected to one positive tab, one side of each positive winding portion in the other part of the positive winding portions is connected to two positive tabs, the two negative tabs of each negative winding portion in the other part of the negative winding portions are arranged axially symmetrically along the thickness direction of the main body portion; and each negative winding portion is connected to one negative tab.

3. The jelly roll battery cell according to claim 1, wherein one side of each negative winding portion in one part of the negative winding portions is connected to one negative tab, one side of each negative winding portion in the other part of the negative winding portions is connected to two negative tabs, and the two negative tabs of each negative winding portion in the other part of the negative winding portions are arranged axially symmetrically along the thickness direction of the main body portion; and each positive winding portion is connected to one positive tab.

4. The jelly roll battery cell according to claim 1, wherein along a direction from inner circles to outer circles, one side of each positive winding portion in the first half of the positive winding portions is connected to two positive tabs, and one side of each positive winding portion in the second half of the positive winding portions is connected to one positive tab; and/or along the direction from inner circles to outer circles, one side of each negative winding portion in the first half of the negative winding portions is connected to two negative tabs, and one side of each negative winding portion in the second half of the negative winding portions is connected to one negative tab.

5. The jelly roll battery cell according to claim 1, wherein along the direction from inner circles to outer circles, one side of each positive winding portion in odd-numbered circles of the positive winding portions is connected to two positive tabs, and one side of each positive winding portion in even-numbered circles of the positive winding portions is connected to one positive tab; alternatively, one side of each positive winding portion in odd-numbered circles of the positive winding portions is connected to one positive tab, and one side of each positive winding portion in even-numbered circles of the positive winding portions is connected to two positive tabs; and/or along the direction from inner circles to outer circles, one side of each negative winding portion in odd-numbered circles of the negative winding portions is connected to two negative tabs, and one side of each negative winding portion in even-numbered circles of the negative winding portions is connected to one negative tab; or alternatively, one side of each negative winding portion in odd-numbered circles of the negative winding portions is connected to one positive tab, and one side of each negative winding portion in even-numbered circles of the negative winding portions is connected to two negative tabs.

6. The jelly roll battery cell according to claim 1, wherein the positive electrode plate comprises a positive current collector and a positive active material layer arranged on one side of the positive current collector; the negative electrode plate comprises a negative current collector and a negative active material layer arranged on one side of the negative current collector; one end of the positive tab is connected to the positive current collector of the corresponding positive winding portion, and the other end thereof is away from the positive current collector along the virtual axis; one end of the negative tab is connected to the negative current collector of the corresponding negative winding portion, and the other end thereof is away from the negative current collector along the virtual axis; and along the thickness direction of the main body portion, the plurality of positive tabs located on the same side of the virtual axis are stacked, and the plurality of negative tabs located on the same side of the virtual axis are stacked.

7. The jelly roll battery cell according to claim 6, wherein the positive tab has the same thickness as the positive current collector; and the negative tab has the same thickness as the negative current collector.
 8. The jelly roll battery cell according to claim 6, wherein in the direction of the virtual axis, the plurality of positive tabs stacked are tabs with an equal length or tabs with gradient lengths; and in the direction of the virtual axis, the plurality of negative tabs stacked are tabs with an equal length or tabs with gradient lengths.
 9. The jelly roll battery cell according to claim 8, wherein the plurality of positive tabs stacked are tabs with an equal length, and the length of the positive tabs is 28%-29% of the width of the positive winding portions connected to the positive tabs; and the plurality of negative tabs stacked are tabs with an equal length, and the length of the negative tabs is 28%-29% of the width of the negative winding portions connected to the negative tabs.
 10. The jelly roll battery cell according to claim 8, wherein the plurality of positive tabs stacked are tabs with gradient lengths; in the plurality of positive tabs, the maximum length of the positive tabs is 28%-29% of the width of the positive winding portions connected to the positive tabs, and the minimum length of the positive tabs is 19%-20% of the width of the positive winding portions connected to the positive tabs; and the lengths of the plurality of positive tabs stacked are decreased progressively layer by layer as per uniform-gradient dimensions along the direction from inner circles to outer circles; the plurality of negative tabs stacked are tabs with gradient lengths; in the plurality of negative tabs, the maximum length of the negative tabs is 28%-29% of the width of the negative winding portions connected to the negative tabs, and the minimum length of the negative tabs is 19%-20% of the width of the negative winding portions connected to the negative tabs; and the lengths of the plurality of negative tabs stacked are decreased progressively layer by layer as per uniform-gradient dimensions along the direction from inner circles to outer circles.
 11. The jelly roll battery cell according to claim 6, wherein the width of the end that is of the positive tab and that is away from the positive current collector is 10%-12.5% of the circumference of the winding portion at the innermost circle of the flat winding body, and the width of the end of the positive tab connected to the positive winding portion is 11%-14% of the circumference of the winding portion at the innermost circle of the flat winding body.
 12. The jelly roll battery cell according to claim 1, wherein the flat winding body has a length direction and a width direction; the virtual axis is parallel to the width direction; and the width of the negative electrode plate is greater than the width of the positive electrode plate, and the length of the negative electrode plate is greater than the length of the positive electrode plate.
 13. The jelly roll battery cell according to claim 1, wherein an innermost circle of the flat winding body is a part of the negative winding portions; and an outermost circle of the flat winding body is a part of the negative winding portions.
 14. The jelly roll battery cell according to claim 1, wherein one side of each positive winding portion in the positive winding portions is connected to one or two positive tabs; one side of each negative winding portion in the negative winding portions is connected to one or two negative tabs; or one side of each positive winding portion in the positive winding portions is connected to one or two positive tabs; in the negative winding portions, the negative winding portion at the innermost circle or the negative winding portion at the outermost circle is not provided with the negative tab; and one side of each of the other negative winding portions is connected to one or two negative tabs.
 15. A battery, comprising the jelly roll battery cell according to claim 1.
 16. A battery assembly, comprising the batteries according to claim 15.
 17. An electric device, comprising the battery assembly according to claim 16.
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